Computer Networks Lecture 1: Core Architectures

Layered Protocol Stack and Transmission Media

A computer network consists of two or more autonomous computing devices linked by communication media to excl
Transmission media are categorized into guided copper or optical fiber and unguided wireless channels.
The physical layer translates logical binary bits into electrical, optical, or radio physical signals.

Computer Networks Lecture 2: Packet Switching and Routing

Datagram Forwarding and IP Addressing

In a packet-switched network, routers inspect destination IP headers and consult forwarding routing tables. Subnet masks partition the 32-bit IPv4 address into network prefixes and individual host identifiers. Hierarchical CIDR routing aggregates multiple subnet prefixes to prevent global routing table explosion.

Computer Networks Lecture 3: Reliable Transport Protocols

TCP Sliding Window and Congestion Avoidance

TCP guarantees ordered reliable delivery using cumulative acknowledgements and retransmission timers.
The sliding window protocol maximizes channel throughput by allowing multiple unacknowledged frames in flight.
Congestion control uses additive-increase multiplicative-decrease AIMD to converge on fair network capacity.